Apibunya "Alpha" Sinworn

Bangkok, Thailand | althanida@gmail.com | +66-96-083-3477
enumaalpha.vercel.app | linkedin.com/in/alpha-sinworn | github.com/Lordalpaca | @a.fun.company

Creative technologist building at the intersection of hardware, software, and art. Designs and ships complete experiences end to end, from interactive installations and live event technology to full-stack AI products.

EXPERIENCE

IT Specialist and Software Developer | Thai Living Law, Bangkok *March - June 2026*
- Built and deployed law firm's full content automation platform end to end as an active commercial tool
- Integrated OpenRouter, Meta Graph API, LINE Messaging API, and Sanity CMS into one dashboard

Product Owner Intern | AIYA, Bangkok *September 2025*
- Researched the competitive landscape and defined product requirements for an AI SaaS platform
- Developed a software interface connecting AI hardware to LINE OA and CRM platforms

Instructional Aide and Teaching Assistant | Arizona State University *2022-2024*
- Designed curriculum and ran labs for Microcontroller Applications, troubleshooting Arduino builds
- Led recitation sessions and created supplementary materials as Signals and Systems Teaching Assistant
- Taught human-centered design and monitored 70+ projects in Engineering Projects in Community Service

SELECTED WORK

The Sparkling Hour: Year 2126 | what a WHY x NEXTOPIA, Siam Paragon *2026*
- Built a scroll-driven cinematic hero site in Next.js, React Three Fiber, GSAP, and canvas scrubbing
- Directed a promo reel remixing Fritz Lang's Metropolis (1927), assembled via an ffmpeg pipeline
- Shipped a live activity streaming attendee submissions onto a projected 3D globe in real time
- Deployed the full stack on Netlify and Supabase behind reverse-proxy routing under one domain

SUKONTHnAI, AI Scent Installation | Bangkok *February 2026*
- Created an installation that reads a visitor's photo with AI and dispenses a custom Thai fragrance
- Architected a Raspberry Pi 5 and Pico system over MQTT mapping AI analysis to 26 essential oils
- Conceived, engineered, and built solo in approximately 52 hours

PKratheoy and KaDealDed | CodeCamp Thailand *2025*
- Built PKratheoy, a mental health journaling app with a mood calendar and cognitive behavioral therapy tools
- Built KaDealDed, a deals platform with auth, QR codes, and an admin dashboard, as Product Owner

NASA L'SPACE Mission Concept Academy | Project Manager *Summer 2024*
- Led a 10-person cross-timezone team designing a lunar rover mission concept
- Managed schedules, milestones, and cross-disciplinary collaboration across time zones

NASA L'SPACE Proposal Writing and Evaluation Experience Academy | Apprentice *Spring 2024*
- Trained in NASA proposal evaluation and review alongside subject matter experts
- Developed working knowledge of NASA project evaluation criteria

Head Movement Cursor Control, Assistive Technology | Arizona State University *Spring 2024*
- Built a cursor control system for people with disabilities using a head-mounted accelerometer
- Designed the hardware-software interface and tuned sensor algorithms for usability and responsiveness

LeafyWear, Wearable Peritoneal Dialysis Device | ASU Capstone *2023-2024*
- Led circuit, motor control, and power design for a device freeing patients from 4-hour clinic sessions
- Conducted stakeholder research with patients, nurses, and physicians to drive user-centered design
- Won 1st Place Popular Vote at Open Pitch and presented at the ASU Biomedical Engineering Symposium

Optogenetic PACE Machine V3, Honors Thesis | ASU Bartelle Lab *2023-2024*
- Designed and fabricated a custom PCB and motor shield for a 5-pump open-source bioreactor
- Iterated the full cycle from breadboard to protoboard to fabricated board, moving from Fusion360 to Altium
- Programmed Arduino automation for media feed control and growth tracking

PCB Silkscreen Artwork, PACE Machine V3 | ASU Bartelle Lab *2023-2024*
- Drew original artwork in Illustrator and embedded it into the silkscreen layer of a working PCB
- Merged art practice and engineering into one object, making a research-grade instrument the canvas

EPICS Lake Litter Solution | Arizona State University *2023-2024*
- Built an autonomous water robot to collect trash at Tempe Town Lake as Electrical and Finance Lead
- Led the pitching team at EPICS Elite Pitch, taking 3rd Place

Space Cow Apocalypse, Board Game | Arizona State University *Spring 2023*
- Directed game pieces, packaging, and visual identity as Art Lead using laser cutting and 3D printing
- Grounded mechanics in user research and statistical analysis of playtesting data

EPICS Coral Reef Restoration, Study Abroad Spain | Arizona State University *Fall 2022*
- Led a team of 7 designing artificial coral reefs for field testing in Spain
- Applied engineering to environmental restoration across a cross-cultural collaboration

EPICS Mental Health Mobile Games | Arizona State University *Fall 2021*
- Designed mental health app UI/UX for the Hopi community with Northern Arizona University
- Created culturally responsive layout and graphics grounded in community input

EDUCATION

Intelligent Automation Certificate | KMITL, Bangkok *2026*
Programmable Logic Controller programming (Mitsubishi/GX Works), ladder logic, industrial automation, and ABB robot programming with path control and sorting on a physical robot arm

Raspberry Pi and AIoT | TESR Academy, Bangkok *January 2026*

ROS 2 Basic | TESR Academy, Bangkok *November 2025*

Full-Stack Developer Trainee | CodeCamp Thailand, Bangkok *2025*
JavaScript, React, Node.js, Express, MySQL, and Git

B.S.E. Biomedical Engineering | Arizona State University, Ira A. Fulton Schools of Engineering *2019-2024*
Barrett, the Honors College | Magna Cum Laude | Dean's List | GPA 3.68/4.00
Honors Thesis: Open Source Hardware for Accessible Directed Evolution Bioreactor

High School Diploma | Triamudom Suksa, Bangkok *2018*

COMMUNITY

Global Shapers Community, World Economic Forum, Bangkok Hub *2025-Present*
- Led Intergen Harmony, a youth-elder mental health initiative in Bangkok
- Delivered it with support from Bangkok Metropolitan Administration and Thai Health Promotion Foundation
- Presented project outcomes to policymakers at parliament

RECOGNITION

- Top 40 Finalist, OKMD Hackathon (2025)
- Finalist, MIT Hacking Medicine x Siriraj (2024)
- 1st Place Popular Vote, Open Pitch, Wearable Dialysis Device, Arizona State University (2024)
- 3rd Place, EPICS Elite Pitch, Arizona State University (2024)

SKILLS

Interactive and Installation: React Three Fiber, WebGL, GSAP, generative AI video and image pipelines, ffmpeg, interactive installation, live event technology
Software: TypeScript, JavaScript, React, Next.js, Python, C/C++, Node.js, Node-RED, N8N, MQTT, Supabase, PostgreSQL, Docker, MATLAB, LabVIEW
Hardware: Raspberry Pi, Arduino, printed circuit board design (Altium, Fusion 360), sensors and actuators, motor control, soldering, 3D printing, Programmable Logic Controller (Mitsubishi), ABB robot arm
Design: Adobe Illustrator, Photoshop, Figma, SOLIDWORKS
AI-augmented development: rapid prototyping and hardware-software integration using AI tools
Languages: Thai (Native), English (Fluent)